

LIKELIHOOD-BASED INFERENCE

We observe a random sample $(y_1, \dots, y_n) = y^{\text{obs}}$

We formulate a statistical model $Y \sim p(y; \theta)$ $\theta \in \Theta$

LIKELIHOOD FUNCTION: the function of θ $p(y^{\text{obs}}; \theta)$, $\theta \in \Theta$, denoted with $L(\theta) = L(\theta; y^{\text{obs}})$

LOG-LIKELIHOOD FUNCTION: the logarithmic transformation of $L(\theta)$, denoted with $\ell(\theta) = \log L(\theta)$

MAXIMUM LIKELIHOOD ESTIMATE: if it exists, is the value $\hat{\theta}$ such that $L(\hat{\theta}) \geq L(\theta)$ for all $\theta \in \Theta$
(equivalently, $\ell(\hat{\theta}) \geq \ell(\theta)$)

before observing the sample, we have the MAXIMUM LIKELIHOOD ESTIMATOR $\hat{\Theta} = \hat{\Theta}(Y)$ (it is a random variable)

SCORE FUNCTION $\ell_*(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$ first derivative

LIKELIHOOD EQUATION $\ell_*(\theta) = 0$

the MLE is a solution of the likelihood equation

HESSIAN $\ell_{**}(\theta) = \frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^T}$ second derivative

example: $Y \sim \text{Pois}(\lambda)$ $p(y; \lambda) = e^{-\lambda} \frac{\lambda^y}{y!}$

I observe $(y_1, y_2) = (3, 4)$

Assuming y is a random sample, $p((y_1, y_2); \lambda) = p(y_1; \lambda) p(y_2; \lambda)$
 $= e^{-\lambda} \frac{\lambda^{y_1}}{y_1!} \cdot e^{-\lambda} \frac{\lambda^{y_2}}{y_2!} = e^{-2\lambda} \frac{\lambda^{y_1+y_2}}{y_1! y_2!}$

likelihood function $L(\lambda) = L(\lambda; (y_1, y_2)) = e^{-2\lambda} \frac{\lambda^7}{3! 4!}$

it is equivalent to consider $K \cdot L(\lambda)$

$$\Rightarrow L(\lambda) = e^{-2\lambda} \lambda^7$$

log-likelihood function $\ell(\lambda) = -2\lambda + 7 \log \lambda$

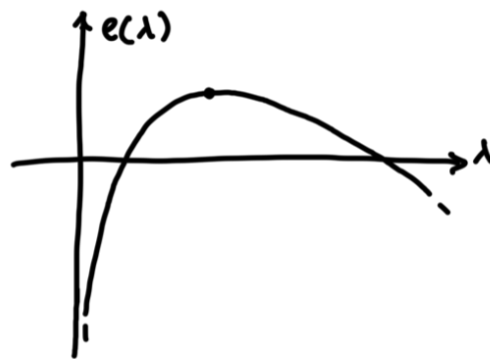
maximum likelihood estimate is $\hat{\lambda} = \arg \max_{\lambda \in (0, \infty)} \ell(\lambda)$

$$\ell'(\lambda) = \frac{d}{d\lambda} \ell(\lambda) = -2 + \frac{7}{\lambda}$$

$$\ell'(\lambda) = 0 \Rightarrow -2 + \frac{7}{\lambda} = 0 \Rightarrow \frac{7}{\lambda} = 2 \Rightarrow \hat{\lambda} = 3.5$$

is it a max?

$$\ell''(\lambda) = -\frac{7}{\lambda^2} < 0 \quad \text{ok!}$$



Before substituting the obs. values: $\ell(\lambda; y) = -2\lambda + (y_1 + y_2) \log \lambda$ and $\hat{\lambda} = \left(\frac{\sum_{i=1}^2 y_i}{2} \right) = \bar{y}$

If I look at $\hat{\lambda}$ as a function of the r.v. $\hat{\Lambda} = \hat{\Lambda}(Y) = \bar{Y} = \frac{1}{2} (Y_1 + Y_2)$

we can study the properties of $\hat{\Lambda}$

$$\mathbb{E}[\hat{\Lambda}] = \mathbb{E}[\bar{Y}] = \mathbb{E}\left[\frac{1}{2} (Y_1 + Y_2)\right] = \frac{1}{2} (\mathbb{E}[Y_1] + \mathbb{E}[Y_2]) = \frac{1}{2} \cdot 2\lambda = \lambda$$

$$\text{var}(\hat{\Lambda}) = \text{var}(\bar{Y}) = \text{var}\left(\frac{1}{2} (Y_1 + Y_2)\right) \stackrel{\text{II}}{=} \frac{1}{4} (\text{var}(Y_1) + \text{var}(Y_2)) = \frac{1}{4} \cdot 2\lambda = \frac{\lambda}{2}$$

LIKELIHOOD RATIO TEST

It is a general procedure to perform tests on nested models (i.e. the simpler model can be obtained starting from the more complex model through constraints on the parameters)

consider a test where both the null and the alternative hypotheses are simple:

$$\begin{cases} H_0: \theta = \theta_0 \\ H_1: \theta = \theta_1 \end{cases}$$

one way to decide between H_0 and H_1 is to compare the likelihood evaluated at θ_0 and θ_1

$$\frac{L(\theta_0; y^{\text{obs}})}{L(\theta_1; y^{\text{obs}})}$$

If θ_0 is better than θ_1 , $L(\theta_0) > L(\theta_1)$. Hence large values of the ratio suggest the acceptance of H_0 .

In general, when the hypotheses are not simple

$$\begin{cases} H_0: \theta \in \Theta_0 \\ H_1: \theta \in \Theta \setminus \Theta_0 \end{cases}$$

The likelihood ratio test is based on the test statistic

$$\begin{aligned} \lambda_{LR} &= -2 \log \left[\frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)} \right] \\ &= -2 [\ell(\theta_0) - \ell(\hat{\theta})] \end{aligned}$$

The exact distribution has to be determined on a case-by-case basis, however

ASYMPTOTIC DISTRIBUTION (holds for large n)

under some regularity conditions, $\lambda_{LR} \sim \chi^2_q$ under H_0

where $q = \dim(\Theta) - \dim(\Theta_0)$